

```
#!/bin/bash

echo "Enter number of processes:"
read n

declare -a pname
declare -a bt
declare -a wt
declare -a tat

total_wt=0
total_tat=0

# Input
for ((i=0; i<n; i++))
do
    echo "Enter Process Name:"
    read pname[$i]
    echo "Enter Burst Time:"
    read bt[$i]
done

# FCFS Calculation
wt[0]=0

for ((i=1; i<n; i++))
do
    wt[$i]=$(( wt[$i-1] + bt[$i-1] ))
done

# Turnaround Time
for ((i=0; i<n; i++))
do
    tat[$i]=$(( wt[$i] + bt[$i] ))
    total_wt=$(( total_wt + wt[$i] ))
    total_tat=$(( total_tat + tat[$i] ))
done

avg_wt=$((echo "scale=2; $total_wt / $n" | bc))

# Output
echo -e "\nProcess\tBurst Time\tWaiting Time\tTurnaround Time"

for ((i=0; i<n; i++))
do
    echo -e "${pname[$i]}\t${bt[$i]}\t\t${wt[$i]}\t\t${tat[$i]}"
done

echo -e "\nTotal Waiting Time: $total_wt"
echo "Average Waiting Time: $avg_wt"
echo "Total Turnaround Time: $total_tat"
```

devyesh@localhost:~/Desktop/devyesh241801049\$./exp5a.sh

Enter number of processes:

2

Enter Process Name:

P1

Enter Burst Time:

5

Enter Process Name:

P2

Enter Burst Time:

3

Process	Burst Time	Waiting Time	Turnaround Time
P1	5	0	5
P2	3	5	8

Total Waiting Time: 5

Average Waiting Time: 2.50

Total Turnaround Time: 13

```

#include <stdio.h>
#include <string.h>

struct process {
    char name[10];
    int at, bt, wt, tat;
};

int main() {
    int n, i, j;
    float total_wt = 0, total_tat = 0;
    struct process p[10], temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("\nEnter Process Name: ");
        scanf("%s", p[i].name);
        printf("Enter Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Enter Burst Time: ");
        scanf("%d", &p[i].bt);

        p[i].wt = 0;
        p[i].tat = 0;
    }

    // Sort based on Burst Time (Ascending)
    for(i = 0; i < n-1; i++) {
        for(j = i+1; j < n; j++) {
            if(p[i].bt > p[j].bt) {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    // Waiting Time Calculation
    p[0].wt = 0;

    for(i = 1; i < n; i++) {
        p[i].wt = p[i-1].wt + p[i-1].bt;
    }

    // Turnaround Time
    for(i = 0; i < n; i++) {
        p[i].tat = p[i].wt + p[i].bt;
        total_wt += p[i].wt;
        total_tat += p[i].tat;
    }

    printf("\nProcess\tAT\tBT\tWT\tTAT\n");

    for(i = 0; i < n; i++) {
        printf("%s\t%d\t%d\t%d\t%d\n",
            p[i].name, p[i].at, p[i].bt,
            p[i].wt, p[i].tat);
    }

    printf("\nAverage Waiting Time = %.2f", total_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", total_tat/n);

    return 0;
}

```

```
devyesh@localhost:~/Desktop/devyesh241801049$ ./exp5b
```

```
Enter number of processes: 3
```

```
Enter Process Name: P1
```

```
Enter Arrival Time: 0
```

```
Enter Burst Time: 6
```

```
Enter Process Name: P2
```

```
Enter Arrival Time: 0
```

```
Enter Burst Time: 4
```

```
Enter Process Name: P3
```

```
Enter Arrival Time: 0
```

```
Enter Burst Time: 2
```

Process	AT	BT	WT	TAT
P3	0	2	0	2
P2	0	4	2	6
P1	0	6	6	12

```
Average Waiting Time = 2.67
```

```
Average Turnaround Time = 6.67
```

```

#include <stdio.h>
#include <string.h>

struct process {
    char name[10];
    int bt, priority, wt, tat;
};

int main() {
    int n, i, j;
    float total_wt = 0, total_tat = 0;
    struct process p[10], temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    // Input
    for(i = 0; i < n; i++) {
        printf("\nEnter Process Name: ");
        scanf("%s", p[i].name);

        printf("Enter Burst Time: ");
        scanf("%d", &p[i].bt);

        printf("Enter Priority: ");
        scanf("%d", &p[i].priority);

        p[i].wt = 0;
        p[i].tat = 0;
    }

    // Sort by Priority (Ascending)
    for(i = 0; i < n-1; i++) {
        for(j = i+1; j < n; j++) {
            if(p[i].priority > p[j].priority) {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    // Waiting Time
    p[0].wt = 0;
    for(i = 1; i < n; i++) {
        p[i].wt = p[i-1].wt + p[i-1].bt;
    }

    // Turnaround Time
    for(i = 0; i < n; i++) {
        p[i].tat = p[i].wt + p[i].bt;
        total_wt += p[i].wt;
        total_tat += p[i].tat;
    }

    // Output
    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for(i = 0; i < n; i++) {
        printf("%s\t%d\t%d\t%d\t%d\n",
            p[i].name, p[i].bt,
            p[i].priority, p[i].wt,
            p[i].tat);
    }

    printf("\nTotal Waiting Time = %.2f", total_wt);
    printf("\nAverage Waiting Time = %.2f", total_wt/n);
    printf("\nTotal Turnaround Time = %.2f\n", total_tat);

    return 0;
}

```

```
devyesh@localhost:~/Desktop/devyesh241801049$ ./exp5c
```

```
Enter number of processes: 2
```

```
Enter Process Name: P1
```

```
Enter Burst Time: 1
```

```
Enter Priority: 3
```

```
Enter Process Name: P2
```

```
Enter Burst Time: 4
```

```
Enter Priority: 1
```

Process	BT	Priority	WT	TAT
P2	4	1	0	4
P1	1	3	4	5

```
Total Waiting Time = 4.00
```

```
Average Waiting Time = 2.00
```

```
Total Turnaround Time = 9.00
```

```

#include <stdio.h>

struct process {
    char name[10];
    int at, bt, wt, tat;
};

int main() {
    int n, i, quantum, t = 0, done;
    float total_wt = 0, total_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    int rem_bt[n];

    printf("Enter Time Quantum: ");
    scanf("%d", &quantum);

    // Input
    for(i = 0; i < n; i++) {
        printf("\nEnter Process Name: ");
        scanf("%s", p[i].name);

        printf("Enter Arrival Time: ");
        scanf("%d", &p[i].at);

        printf("Enter Burst Time: ");
        scanf("%d", &p[i].bt);

        rem_bt[i] = p[i].bt;
        p[i].wt = 0;
    }

    // Round Robin Scheduling
    do {
        done = 1;

        for(i = 0; i < n; i++) {
            if(rem_bt[i] > 0) {
                done = 0;

                if(rem_bt[i] > quantum) {
                    t += quantum;
                    rem_bt[i] -= quantum;
                }
                else {
                    t += rem_bt[i];
                    p[i].wt = t - p[i].bt;
                    rem_bt[i] = 0;
                }
            }
        }
    } while(!done);

    // Turnaround Time
    for(i = 0; i < n; i++) {
        p[i].tat = p[i].bt + p[i].wt;
        total_wt += p[i].wt;
        total_tat += p[i].tat;
    }

    // Output
    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i = 0; i < n; i++) {
        printf("%s\t%d\t%d\t%d\n",
            p[i].name, p[i].bt,
            p[i].wt, p[i].tat);
    }

    printf("\nAverage Waiting Time = %.2f", total_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", total_tat/n);

    return 0;
}

```

```
devyesh@localhost:~/Desktop/devyesh241801049$ ./exp5d
```

```
Enter number of processes: 2
```

```
Enter Time Quantum: 2
```

```
Enter Process Name: P1
```

```
Enter Arrival Time: 0
```

```
Enter Burst Time: 5
```

```
Enter Process Name: P2
```

```
Enter Arrival Time: 0
```

```
Enter Burst Time: 3
```

Process	BT	WT	TAT
P1	5	3	8
P2	3	4	7

```
Average Waiting Time = 3.50
```

```
Average Turnaround Time = 7.50
```